P29 CBDC Project Presentation

Team: P29

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What We'll Cover Today:

- Project Overview & Objectives
- Key Achievements
- Next Steps

What We Built:

- A complete Central Bank Digital Currency system for Australia
- Permissioned blockchain network using Hyperledger Fabric

Organization	Role
RBA	Central Bank - Controls monetary policy
BankA	Commercial Bank - Customer accounts
BankB	Commercial Bank - Customer accounts
AUSTRAC	Regulator - Transaction monitoring

Key Achievements

1. Blockchain Network

-All 4 organisations running

```
CONTAINER ID   IMAGE
                                                                  COMMAND                  CREATED         STATUS
 PORTS
                 NAMES
7307c6afa381   dev-peer0.org2.example.com-eaud_1.0-701ba373012eeaf2b13700a35cbf9816913094df80cec5927f86744e2a729af7-e223
6b5b30d8765e4b9af13db174860d69160dfbfb71681588ccccf2e9d61d8a   "docker-entrypoint.s…"   39 minutes ago   Up 39 minutes

                 dev-peer0.org2.example.com-eaud_1.0-701ba373012eeaf2b13700a35cbf9816913094df80cec5927f86744e2a729af7
1e576b042968   dev-peer0.org1.example.com-eaud_1.0-701ba373012eeaf2b13700a35cbf9816913094df80cec5927f86744e2a729af7-cdcb
bad7fc5144205c3f54684b469c57b51fa191061379e183336d6ed4970865   "docker-entrypoint.s…"   39 minutes ago   Up 39 minutes

                 dev-peer0.org1.example.com-eaud_1.0-701ba373012eeaf2b13700a35cbf9816913094df80cec5927f86744e2a729af7
a70d8ad199c0   hyperledger/fabric-peer:latest
                                                                  "peer node start"        3 days ago       Up 39 minutes
 0.0.0.0:7051->7051/tcp, [::]:7051->7051/tcp, 0.0.0.0:9444->9444/tcp, [::]:9444->9444/tcp
                 peer0.org1.example.com
be676897480c   hyperledger/fabric-peer:latest
                                                                  "peer node start"        3 days ago       Up 39 minutes
 0.0.0.0:9051->9051/tcp, [::]:9051->9051/tcp, 0.0.0.0:9445->9445/tcp, [::]:9445->9445/tcp
                 peer0.org2.example.com
52d765389d91   hyperledger/fabric-orderer:latest
                                                                  "orderer"                3 days ago       Up 39 minutes
 0.0.0.0:7050->7050/tcp, [::]:7050->7050/tcp, 0.0.0.0:7053->7053/tcp, [::]:7053->7053/tcp, 0.0.0.0:9443->9443/tcp, [::]:
9443->9443/tcp   orderer.example.com
c5d14145ff97   couchdb:3.4.2
                                                                  "tini -- /docker-ent…"   3 days ago       Up 39 minutes
 0.0.0.0:7984->5984/tcp, [::]:7984->5984/tcp
                 couchdb1
c53bfa465093   couchdb:3.4.2
                                                                  "tini -- /docker-ent…"   3 days ago       Up 39 minutes
```

eAUD chaincode deployed on all peers

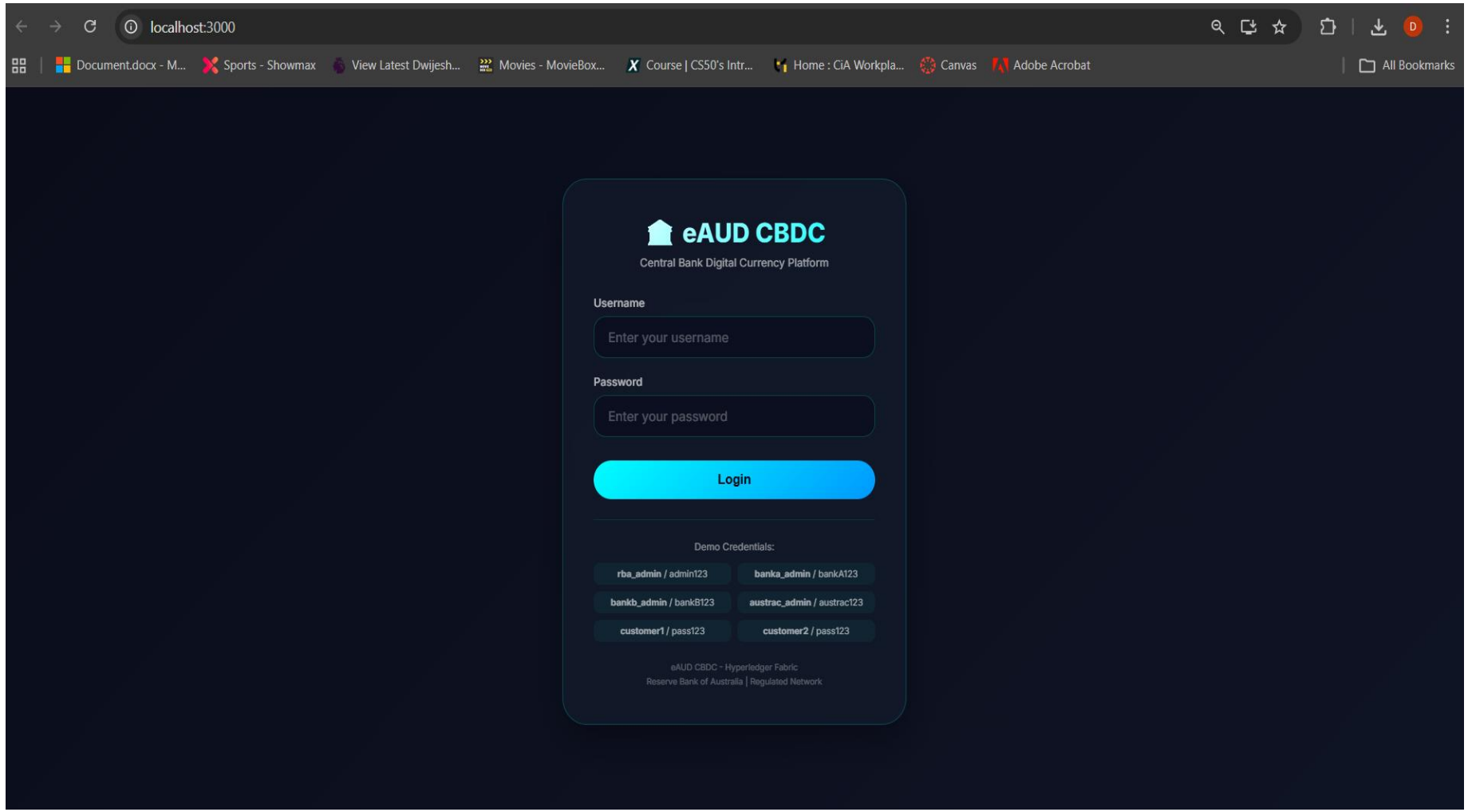
```
Creating channel eaudchannel
Adding orderers
+ . scripts/orderer.sh eaudchannel
+ '[' 0 -eq 1 ']'
+ res=0
2026-03-29 22:29:20.715 AEDT 0001 INFO [channelCmd] InitCmdFactory -> Endorser and orderer connections initialized
Error: proposal failed (err: bad proposal response 500: cannot create ledger from genesis block: ledger [eaudchannel] al
ready exists with state [ACTIVE])
Status: 405
{
        "error": "cannot join: channel already exists"
}

Channel 'eaudchannel' created
Joining org1 peer to the channel...
Using organization 1
+ peer channel join -b ./channel-artifacts/eaudchannel.block
+ res=1
+ peer channel join -b ./channel-artifacts/eaudchannel.block
+ res=1
+ peer channel join -b ./channel-artifacts/eaudchannel.block
+ res=1
+ peer channel join -b ./channel-artifacts/eaudchannel.block
+ res=1
```

chaincode functions (CreateWallet, AddFunds, TransferEAUD, QueryWallet)

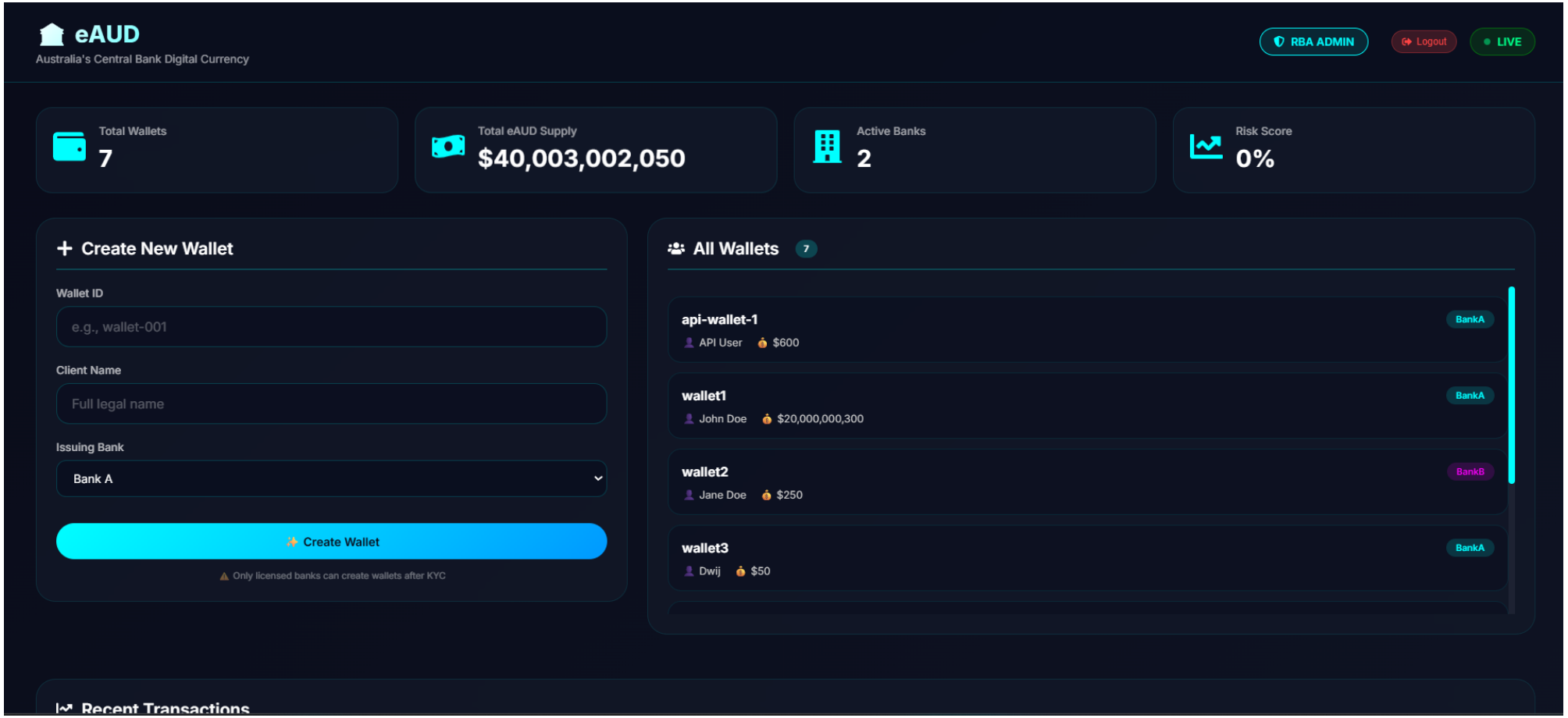
```
dwijesh@Dwijesh:~/fabric-samples/test-network$ peer chaincode invoke -o localhost:7050 --ordererTLSHostnameOverride orderer.exa
mple.com --tls --cafile ${PWD}/organizations/ordererOrganizations/example.com/tlsca/tlsca.example.com-cert.pem -C eaudchannel -
n eaud -c '{"function":"CreateWallet","Args":["wallet1","John Doe","BankA"]}'
2026-03-26 14:33:55.185 AEDT 0001 INFO [chaincodeCmd] chaincodeInvokeOrQuery -> Chaincode invoke successful. result: status:200
 payload:"{\"walletId\":\"wallet1\",\"clientName\":\"John Doe\",\"bankId\":\"BankA\",\"balance\":0,\"createdAt\":\"2026-03-26T0
3:33:55.175Z\",\"status\":\"ACTIVE\"}"
dwijesh@Dwijesh:~/fabric-samples/test-network$ peer chaincode query -C eaudchannel -n eaud -c '{"function":"QueryWallet","Args"
:["wallet1"]}'
{"walletId":"wallet1","clientName":"John Doe","bankId":"BankA","balance":0,"status":"ACTIVE"}
dwijesh@Dwijesh:~/fabric-samples/test-network$ peer chaincode invoke -o localhost:7050 --ordererTLSHostnameOverride orderer.exa
mple.com --tls --cafile ${PWD}/organizations/ordererOrganizations/example.com/tlsca/tlsca.example.com-cert.pem -C eaudchannel -
n eaud -c '{"function":"AddFunds","Args":["wallet1","1000"]}'
2026-03-26 14:34:18.979 AEDT 0001 INFO [chaincodeCmd] chaincodeInvokeOrQuery -> Chaincode invoke successful. result: status:200
 payload:"{\"walletId\":\"wallet1\",\"newBalance\":1000}"
dwijesh@Dwijesh:~/fabric-samples/test-network$ peer chaincode query -C eaudchannel -n eaud -c '{"function":"QueryWallet","Args"
:["wallet1"]}'
{"walletId":"wallet1","clientName":"John Doe","bankId":"BankA","balance":1000,"status":"ACTIVE"}
dwijesh@Dwijesh:~/fabric-samples/test-network$ peer chaincode invoke -o localhost:7050 --ordererTLSHostnameOverride orderer.exa
mple.com --tls --cafile ${PWD}/organizations/ordererOrganizations/example.com/tlsca/tlsca.example.com-cert.pem -C eaudchannel -
n eaud -c '{"function":"CreateWallet","Args":["wallet2","Jane Doe","BankB"]}'
2026-03-26 14:34:28.869 AEDT 0001 INFO [chaincodeCmd] chaincodeInvokeOrQuery -> Chaincode invoke successful. result: status:200
 payload:"{\"walletId\":\"wallet2\",\"clientName\":\"Jane Doe\",\"bankId\":\"BankB\",\"balance\":0,\"createdAt\":\"2026-03-26T0
3:34:28.865Z\",\"status\":\"ACTIVE\"}"
dwijesh@Dwijesh:~/fabric-samples/test-network$ peer chaincode invoke -o localhost:7050 --ordererTLSHostnameOverride orderer.exa
mple.com --tls --cafile ${PWD}/organizations/ordererOrganizations/example.com/tlsca/tlsca.example.com-cert.pem -C eaudchannel -
n eaud -c '{"function":"TransferEAUD","Args":["wallet1","wallet2","300"]}'
2026-03-26 14:34:39.017 AEDT 0001 INFO [chaincodeCmd] chaincodeInvokeOrQuery -> Chaincode invoke successful. result: status:200
 payload:"{\"fromWallet\":\"wallet1\",\"toWallet\":\"wallet2\",\"amount\":300,\"status\":\"success\"}"
dwijesh@Dwijesh:~/fabric-samples/test-network$ peer chaincode query -C eaudchannel -n eaud -c '{"function":"QueryWallet","Args"
:["wallet1"]}'
peer chaincode query -C eaudchannel -n eaud -c '{"function":"QueryWallet","Args":["wallet2"]}'
{"walletId":"wallet1","clientName":"John Doe","bankId":"BankA","balance":700,"status":"ACTIVE"}
{"walletId":"wallet2","clientName":"Jane Doe","bankId":"BankB","balance":300,"status":"ACTIVE"}
dwijesh@Dwijesh:~/fabric-samples/test-network$
```

2. Frontend Dashboard

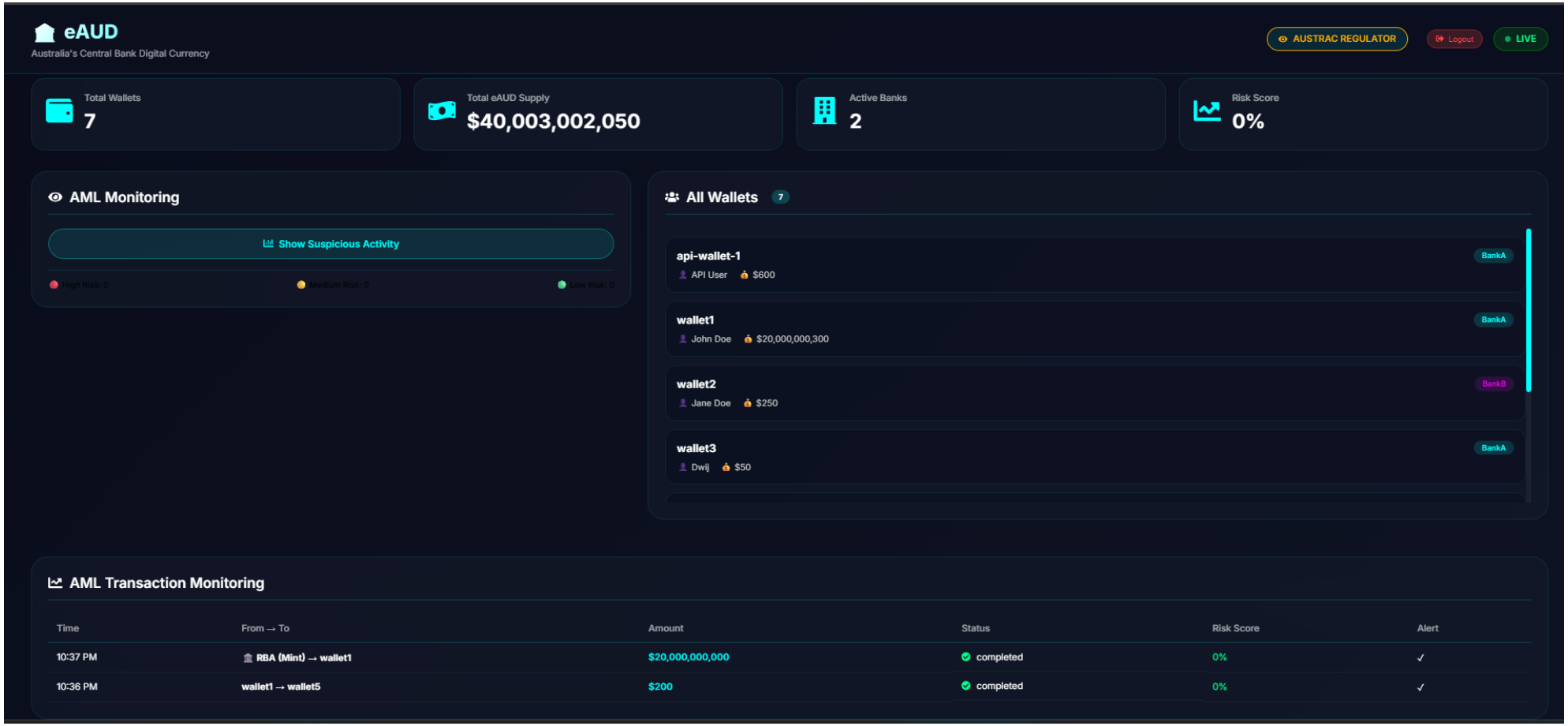
- Login page with demo credentials visible



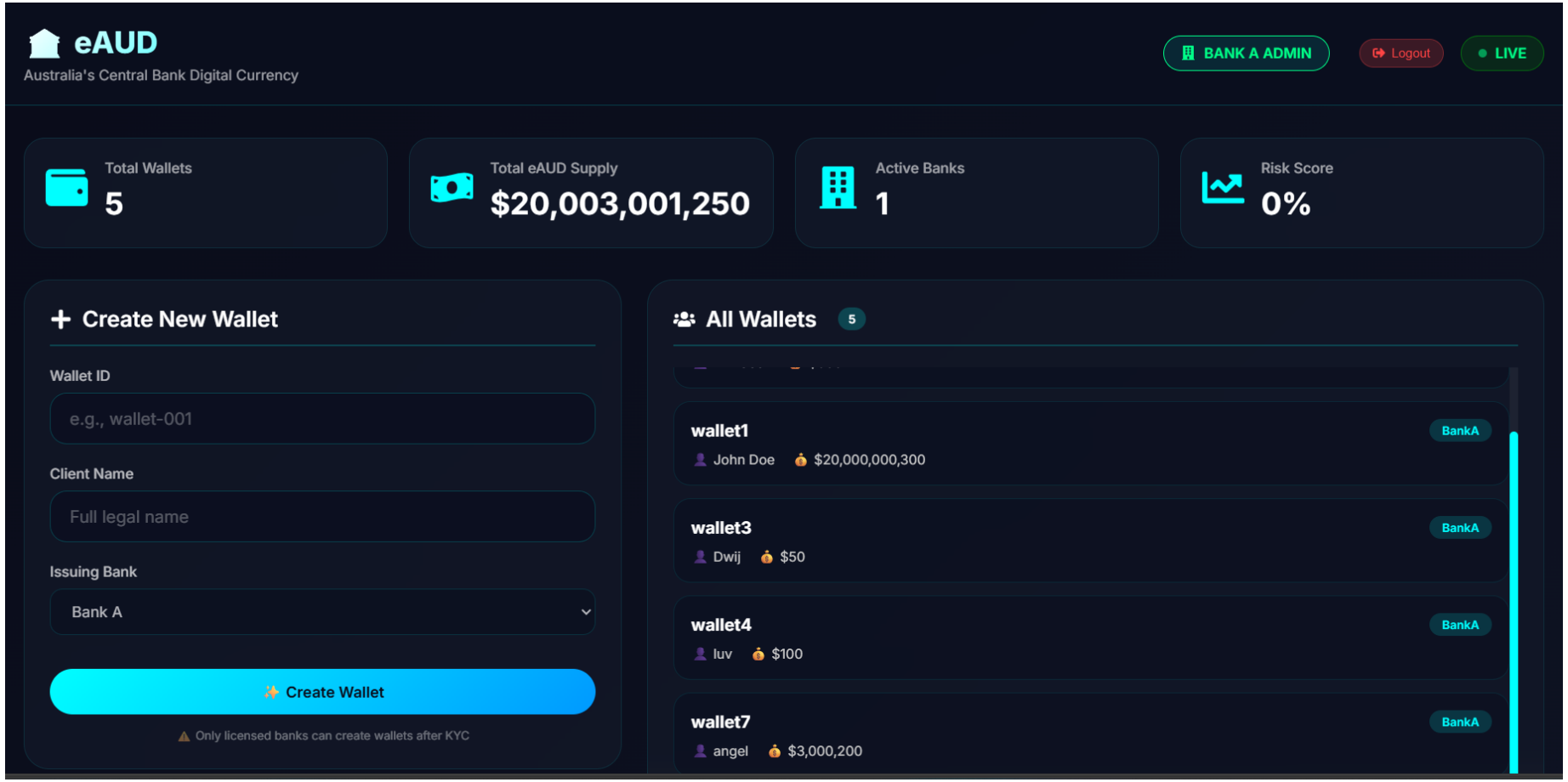
RBA Admin dashboard



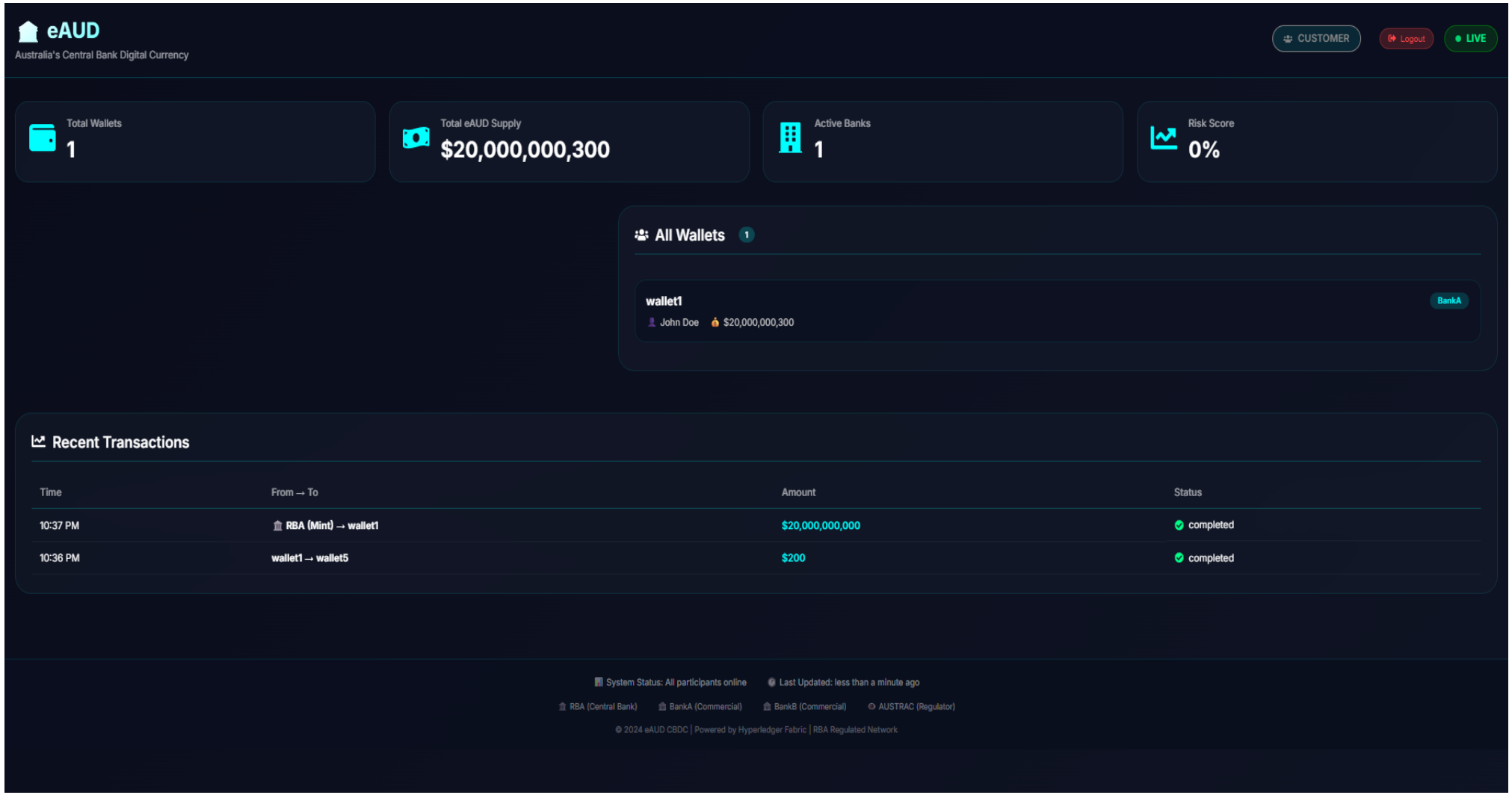
AUSTRAC dashboard with transaction monitoring and risk scores



Bank A Admin dashboard

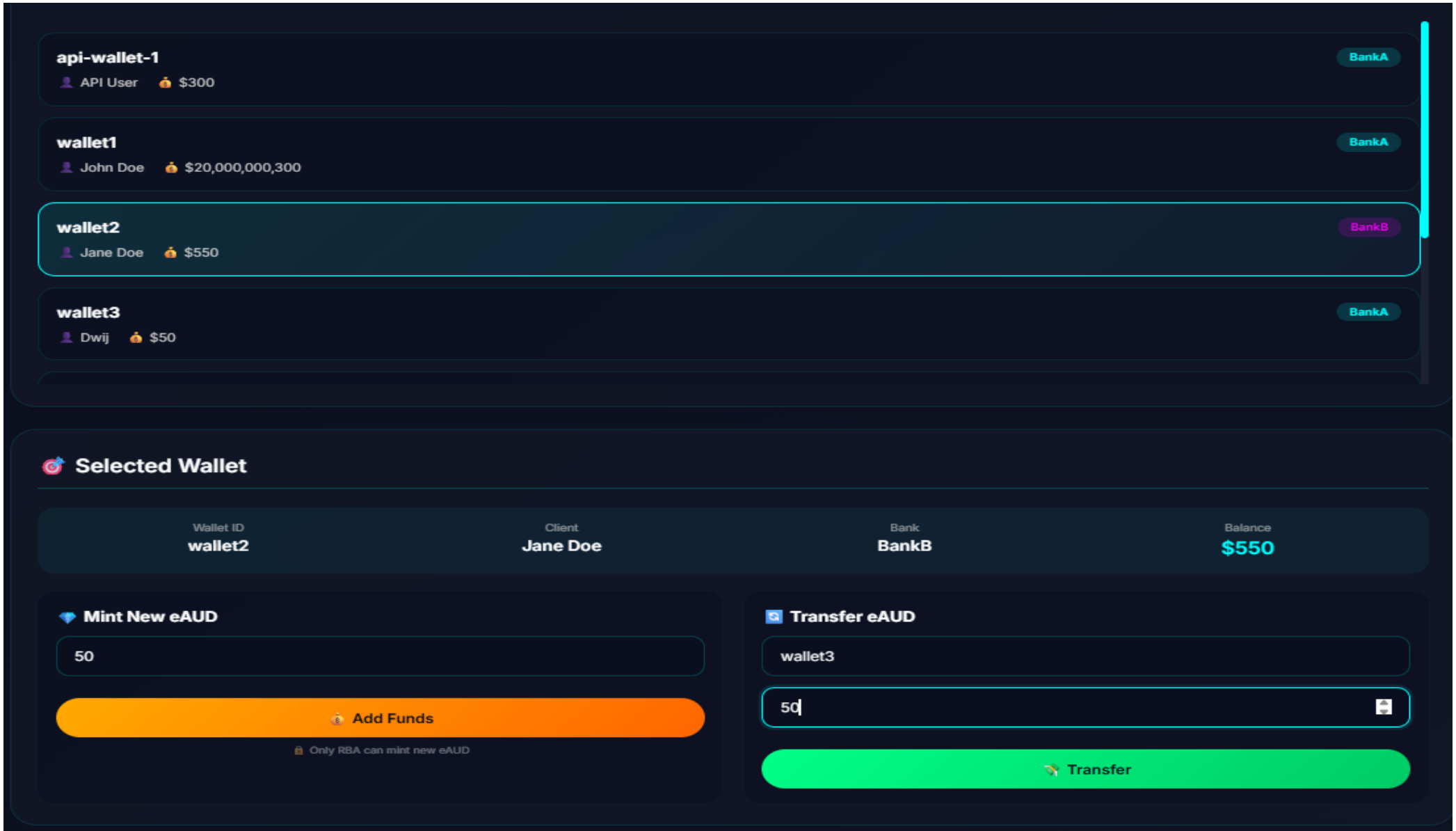


Customer wallet view (single wallet only)

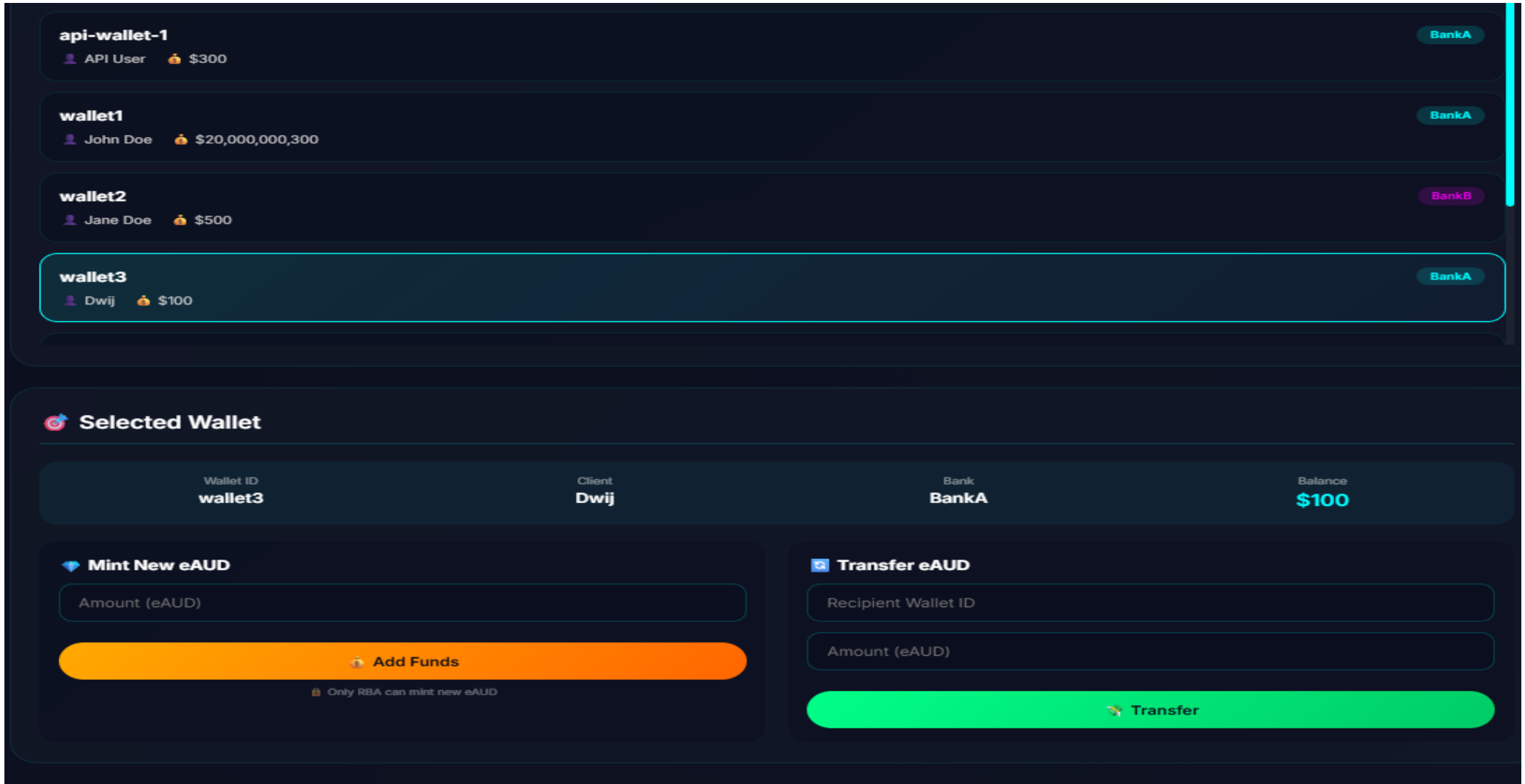


Transactions

- Transferring $50 from wallet2 to wallet3



Result


api-wallet-1
API User
$300
BankA
wallet1
John Doe
$20,000,000,300
BankA
wallet2
Jane Doe
$500
BankB
wallet3
Dwij
$100
BankA
Selected Wallet
Wallet ID
wallet3
Client
Dwij
Bank
BankA
Balance
$100
Mint New eAUD
Amount (eAUD)
Add Funds
Only RBA can mint new eAUD
Transfer eAUD
Recipient Wallet ID
Amount (eAUD)
Transfer

Next Steps

- Add transaction charts and analytics graphs
- Implement PDF export for AML reports
- Dockerize the entire application for easy deployment